

Lecture 3 — OSI and TCP/IP Models

Post-lecture notes for students

The one idea from this lecture. A network is far too complicated to design, build or debug as a single thing. Layering is the trick that makes it tractable: each layer does one job, speaks only to the layers directly above and below it, and deliberately knows nothing about the rest. Every topic for the remainder of this course sits inside one of these layers.

0. Where we left off

Lecture 2 established four results, and this lecture answers the question they leave open.

From Lecture 2	What it said
Packet > circuit switching	Data is split into chunks that share links dynamically. Far more efficient for bursty traffic than reserving idle, dedicated paths.
The store-and-forward rule	Routers must fully receive, verify and buffer an entire packet before pushing the first bit onward to the next hop.
The anatomy of delay	Every packet pays processing, queuing, transmission (pushing bits) and propagation (physical travel) delay.
Throughput and bottlenecks	Your end-to-end speed is not your ISP plan. It is capped by the single slowest bottleneck link on the whole path.

So the core moves packets, and we can now reason about how fast. What we have not addressed is the harder question: thousands of different networks, devices, operating systems and applications, built by companies that never spoke to one another, somehow agree on the format of those packets and the order of operations. How?

1. Objective of today's class

How data travels across the network — not in terms of wires and delay this time, but in terms of the structure of the conversation itself.

2. The idea behind layering

2.1 Sending a letter

You write a letter, put it in an envelope, and hand it to the postal service.

Does the mail carrier care about the letter's contents? No.

And that "no" is the whole idea. The carrier needs the address on the outside of the envelope and nothing else. You need to know how to write a letter and nothing about sorting offices, vans or flight routes. Neither of you needs to understand the other's job, and the letter still arrives.

AND WHAT IF THEY DO CARE?

This was the question posed in class, and it is worth sitting with. If the carrier reads the letter, the separation breaks: your privacy is gone, and the carrier now has to understand every kind of message anybody might send. A layer that looks inside the layer above it stops being a clean layer.

Keep this in mind. It is the seed of two later topics: **encryption** (making the contents unreadable to the carrier) and the question of **which devices are allowed to look how deep** into your traffic.

2.2 What layering buys us

- Each layer solves one problem, and can be understood on its own.
- A layer can be replaced without disturbing the others — swap WiFi for Ethernet and nothing above needs rewriting.
- Different vendors can build different layers and still interoperate, because the boundary between layers is agreed in advance.
- When something breaks, you can localise the fault to a layer instead of searching the whole system.

3. Encapsulation

3.1 The nesting doll

Encapsulation works like a set of nesting dolls — the toy shown in class.

- **Going down the sender's stack:** each layer wraps the data handed to it from above with its own header.
- **Coming up the receiver's stack:** decapsulation reverses the process. Each layer strips off its own specific header and hands the remaining payload upward.

The important property is symmetry. Layer 3 on the receiving machine reads exactly the header that layer 3 on the sending machine wrote, and neither of them touches anything belonging to another layer. Each layer is, in effect, having a private conversation with its counterpart at the other end.

3.2 The gift-wrapping version

The same idea, told the way it was framed in class for the OSI model: you wrap a gift in layers — paper, then a box, then ribbon — going out, and you unwrap them in exactly the reverse order when the gift is received. You cannot untie the ribbon last.

WHY THIS MATTERS MORE THAN IT LOOKS

Encapsulation is the mechanism that makes layering real rather than merely a diagram. It is also why a packet captured off a real network looks like a series of nested headers followed, eventually, by your actual data. When we start reading real captures later in the course, you will be peeling exactly these layers.

4. The OSI model

4.1 What it is

- OSI stands for Open Systems Interconnection.
- It divides network communication into seven layers, in order to simplify the process of designing and reasoning about it.
- It was developed by the ISO — the International Organization for Standardization.

4.2 Why learn the OSI model?

Reason	Why it holds
Everything you use runs on it	WhatsApp, games, Netflix — all of it. Learn the model once and the internet stops being magic.
One map for a messy problem	It splits the impossibly broad task of "send data across the world" into seven clean, separable layers.
It is the industry's shared language	When a colleague says "that's a Layer 4 issue", you will know what they mean — and you will be able to say it yourself, in an interview or a debugging session.
It is the foundation for everything after	Cybersecurity, cloud, AI infrastructure, backend engineering: all of it is built on these layers.

4.3 Importance

- It provides a standard for network communication.
- It ensures interoperability between different systems and devices — machines built by companies that never coordinated can still talk.

A MEMORY AID

Bottom-up (Layer 1 → 7): *Please Do Not Throw Sausage Pizza Away* — Physical, Data Link, Network, Transport, Session, Presentation, Application.

Top-down (Layer 7 → 1): *All People Seem To Need Data Processing* — Application, Presentation, Session, Transport, Network, Data Link, Physical.

Learn the numbers as well as the names. In practice people say "Layer 3" far more often than they say "Network layer".

5. The seven layers, one at a time

We worked down the stack, from the layer closest to you to the layer closest to the physical world.

5.1 Layer 7 — Application

What it does. The Application layer delivers network services directly to end users and supports applications such as web browsers, email and file transfer. Its unit of concern is simply data.

The analogy. A waiter in a restaurant. It takes your order (the user request), relays it to the kitchen (the network), and comes back with your food (the data). The customer never enters the kitchen; the chef never speaks to the customer. The waiter's whole job is connecting the user to the service efficiently.

5.2 Layer 6 — Presentation

What it does. The Presentation layer formats, encrypts and compresses data so that it is readable and secure for the receiving application. Two headline responsibilities: translation and encryption.

Concrete jobs that live here:

- ASCII text to Unicode, and other character-encoding conversions.
- Compression and decompression — zip and unzip.
- Encryption and decryption.
- Encoding images according to their type: PNG, JPEG, GIF.

The analogy. A translator who converts, formats and secures messages so that sender and receiver understand each other even when they speak different languages. Framed another way from the slides: the data has to be presented the way the application wants it, and the way your system wants it — those are not always the same thing.

5.3 Layer 5 — Session

What it does. The Session layer establishes, manages and terminates communication sessions between two devices. Its two named responsibilities are dialog control and synchronization.

The analogy. A courtroom — the judge calling "Order! Order!" and keeping the proceedings continuous. Or, equivalently, a moderator on a panel: it starts the call, keeps the connection running smoothly, and ends it when everyone is finished. Somebody has to decide whose turn it is and when the conversation is over.

5.4 Layer 4 — Transport

What it does. The Transport layer ensures reliable and efficient delivery of data between two devices across a network. A message is encapsulated with detailed source and destination addressing, broken up for transit, and reassembled at the receiver.

The analogy. Imagine many separate conversations sharing one phone line. Multiplexing is what ensures each message reaches the right recipient rather than the wrong one.

THE LINK BACK TO LECTURE 2

This is where TCP lives — the protocol whose job, as we said last lecture, is to paper over the fact that the core offers no guarantees. Loss detection, retransmission and pacing are all Layer 4 concerns. When we reach TCP in Weeks 5–6, we will be working entirely inside this layer.

5.5 Layer 3 — Network

What it does. The Network layer is responsible for logical addressing and for forwarding data packets across different networks. Its job, stated plainly: get packets from point A to point B.

The analogy. A GPS system, finding the best route for delivery trucks — the packets — to reach their destination. It handles addressing and routing, much as you would choose the best roads to arrive fastest and most safely.

Note the word "different" in "across different networks". Layer 3 is the layer that stops caring whether the next network is Ethernet, WiFi or fibre. That indifference is what allows a network of networks to exist at all.

5.6 Layer 2 — Data Link

What it does. The Data Link layer ensures reliable data transfer between two directly connected nodes, and handles error detection and MAC addressing. Its scope is deliberately narrow: one hop, done correctly.

Two responsibilities named in class:

- **Media access control** — "one at a time!" When several devices share a medium, something must decide who transmits when, or they will talk over each other.
- **Error checking.** Detecting that the bits which arrived are the bits that were sent.

Contrast Layer 2 with Layer 3 carefully, because this is the distinction most often confused: Layer 3 worries about the whole journey; Layer 2 worries only about getting safely to the other end of one link.

5.7 Layer 1 — Physical

What it does. The Physical layer transmits raw bits — zeros and ones — over a physical medium.

The media discussed in class:

- Coaxial cable
- Fibre optic cable
- Radio waves

The analogy. The roads, wires and signals the postman actually uses to move around. No addresses, no routes, no meaning — just the physical means of getting something from here to there.

This is also the layer where Lecture 2's propagation delay lives. The speed of light in fibre is a Layer 1 fact, and no amount of clever protocol design at any higher layer can argue with it.

5.8 The seven layers at a glance

#	Layer	One-line job	Analogy from class
7	Application	Delivers network services to the user and their apps	The waiter
6	Presentation	Translates, compresses and encrypts	The translator
5	Session	Starts, maintains and ends the conversation	The judge / panel moderator
4	Transport	Reliable, efficient delivery; multiplexing	Many calls on one phone line
3	Network	Logical addressing and routing between networks	The GPS
2	Data Link	Reliable transfer over one direct link; MAC; error checks	One at a time, please
1	Physical	Moves raw bits over a medium	The roads and wires

6. The TCP/IP model

6.1 Introduction

The TCP/IP model is a layered framework that defines how data is transmitted over the internet, using standard protocols such as TCP and IP.

The important thing to understand about it is its status. OSI is the model we use to think and to teach. TCP/IP is the model the internet actually runs on. They are not competitors so much as two different kinds of object — one a clean conceptual scheme, the other a working system that grew.

6.2 OSI compared with TCP/IP

#	OSI model	TCP/IP model
1	Open Systems Interconnection	Transmission Control Protocol / Internet Protocol
2	7 different layers	4 different layers
3	Developed by ISO (International Organization for Standardization)	Developed by DARPA (Defense Advanced Research Projects Agency)
4	A conceptual model	Used in actual data transmission between computers